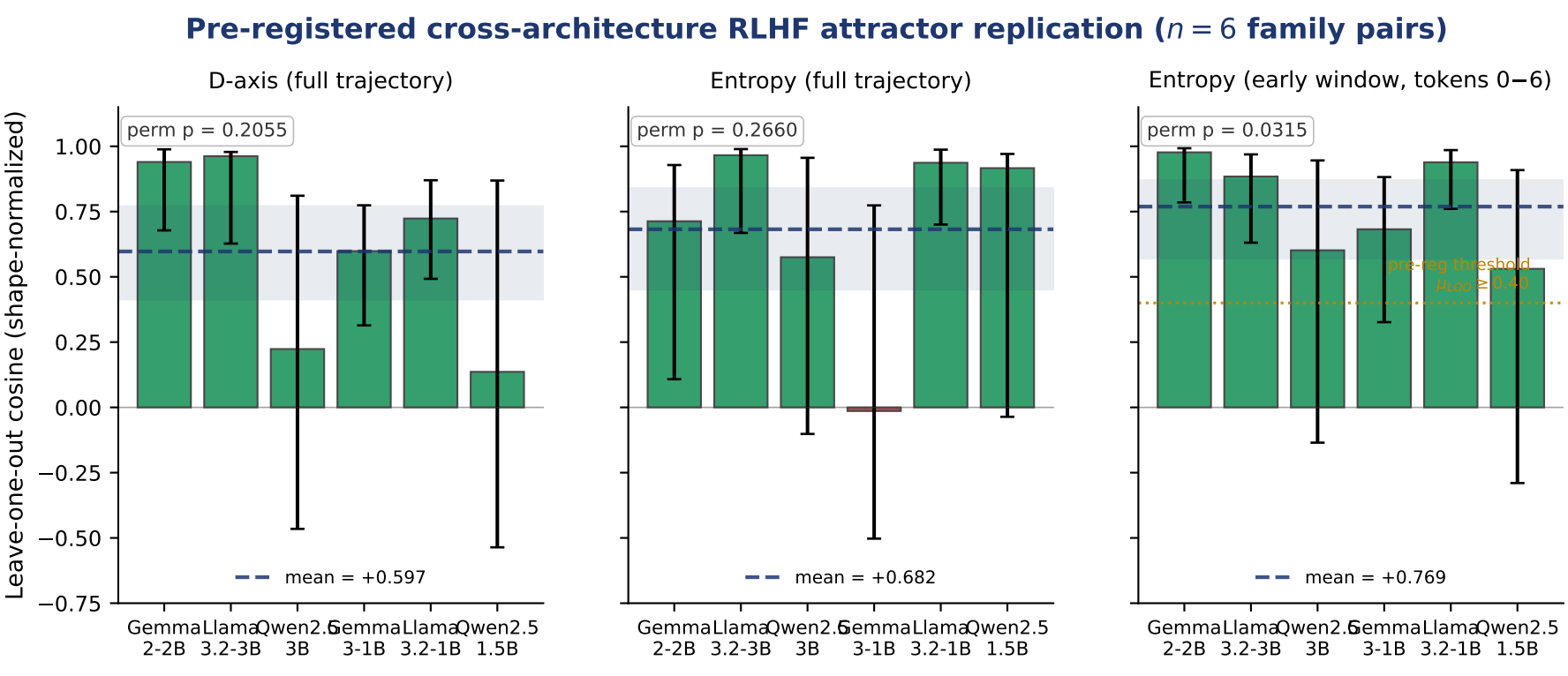
Pre-registered cross-architecture RLHF attractor replication (n = 6 family pairs)
D-axis (full trajectory)
Entropy (full trajectory)
Entropy (early window, tokens 0−6)
Leave-one-out cosine (shape-normalized)
perm p = 0.2055
perm p = 0.2660
perm p = 0.0315
mean = +0.597
mean = +0.682
mean = +0.769
pre-reg threshold
μLOO ≥ 0.40
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Gemma 2-2B
Llama 3.2-3B
Qwen2.5 3B
Gemma 3-1B
Llama 3.2-1B
Qwen2.5 1.5B